



Determinants and performance outcomes of artificial intelligence adoption: Evidence from U.S. Hospitals

Phuoc Pham^{a,*}, Huilan Zhang^b, Wenlian Gao^c, Xiaowei Zhu^a

^a Department of Management, West Chester University, West Chester, PA 19383, USA

^b Department of Accounting, Pennsylvania State University at Altoona, Altoona, PA 16601, USA

^c Department of Finance, Northern Illinois University, DeKalb, IL 60115, USA

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ABSTRACT

Integrating Artificial Intelligence (AI) technology in hospitals offers a unique opportunity to improve hospitals' operating and financial performance. This study is among the first to investigate the determinants and subsequent performance outcomes associated with AI adoption. Using an extensive dataset encompassing 941 AI hospital-year observations and 941 non-AI hospital-year observations, we find that hospitals with a larger market share are great candidates to adopt AI. Furthermore, these hospitals can leverage AI technology to enhance various aspects of performance, including total outpatient revenue, total inpatient revenue, productivity, and occupancy. Importantly, we demonstrate that controlling for endogeneity is essential in assessing the performance outcomes of AI adoption. Our findings shed light on the determinants of AI adoption decisions in healthcare and underscore the manifold benefits AI technology brings to hospital operations and financial outcomes.

1. Introduction

Artificial Intelligence (AI) and data analytics are increasingly prevalent in healthcare since these technologies have been approved as effective tools for helping not only patients but also physicians and healthcare staff. These technologies can help patients by reducing waiting times and making healthcare more affordable. They also help physicians to provide more accurate diagnostics and reduce burnout from paperwork and administrative tasks. According to Accenture's (2020) analysis, key clinical health AI applications can potentially create \$150 billion in annual savings for the United States healthcare economy by 2026.

Nuance (2018) reports a case study of its Computer-Assisted Physician Documentation™ (CAPD) at Universal Health Services, Inc. (UHS), a Fortune 500 company with 26 acute care hospitals in the U.S. and over 200 behavioral health facilities in the U.S. and England. UHS has over 70,000 employees and a combined annual revenue of \$10 billion. CAPD, an intuitive cloud-based speech, and Clinical Documentation Improvement (CDI) workflow help UHS to engage physicians in quality improvement. CAPD tackles the challenges of engaging independent physicians in documentation-improvement efforts, reducing transcription costs, increasing documentation speeds, and capturing more

accurate and detailed records of patient conditions to improve quality measures and reimbursement. In particular, this project achieves a 69 % reduction in transcription costs, netting \$3 million in savings annually; gains a 12 % increase in the case mix index (CMI), which is used in determining the allocation of resources to treat patients; improves documentation of patients with extreme severity of illness by 36 %; and expands the details recorded on patients with the highest risk of mortality by 24 %.

While academic and industrial interest in AI applications in healthcare is growing, there is no consensus on whether AI adoption influences hospital performance (Raimo et al., 2022). Recognizing the promising AI benefits, the literature also highlights barriers to AI adoption, including the cost of AI infrastructure and implementation (Kraus et al., 2021), in-house AI expertise (Dicunzo et al., 2022), and data challenges (Sun & Medaglia, 2019), among others. These factors can diminish returns on hospitals' AI investments, thus reducing the rate of AI adoption. Given the scarcity of rigorous empirical research in this field (Choudhury & Asan, 2020), it is critically important to provide statistical evidence of whether AI adoption improves hospitals' efficiency and effectiveness (Coccia, 2020; Secinaro et al., 2021).

This study aims to fill gaps in the literature by investigating the determinants of AI adoption in U.S. hospitals and the impact of AI adoption

* Corresponding author.

E-mail addresses: papham@wcupa.edu (P. Pham), hxz148@psu.edu (H. Zhang), wgao@niu.edu (W. Gao), xzhu@wcupa.edu (X. Zhu).

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